

torch.linalg.ldl_factor_ex#

https://pytorch.org/docs/stable/generated/torch.linalg.ldl_factor_ex.html

Description:

This is a version of `ldl_factor()` that does not perform error checks unless `check_errors= True`. It also returns the info tensor returned by LAPACK's `sytrf`. info stores integer error codes from the backend library. A positive integer indicates the diagonal element of DDD that is zero. Division by 0 will occur if the result is used for solving a system of linear equations. info filled with zeros indicates that the factorization was successful. If `check_errors=True` and info contains positive integers, then a `RuntimeError` is thrown. When the inputs are on a CUDA device, this function synchronizes only when `check_errors= True`. This function is “experimental” and it may change in a future PyTorch release.

Function Signature

```
torch.linalg.ldl_factor_ex(A, *, hermitian=False,  
check_errors=False, out=None)#
```

Parameters

Keyword Arguments

Returns

Parameters

Keyword

`hermitian` (bool, optional) – whether to consider the input to be Hermitian or symmetric. For real-valued matrices, this switch has no effect. Default: False.

`check_errors` (bool, optional) – controls whether to check the content of `info` and raise an error if it is non-zero. Default: False.

`out` (tuple, optional) – tuple of three tensors to write the output to. Ignored if None. Default: None.

Returns:

A named tuple (LD, pivots, info).

Code Examples

```
>>> A = torch.randn(3, 3)
>>> A = A @ A.mT # make symmetric
>>> A
tensor([[7.2079, 4.2414, 1.9428],
        [4.2414, 3.4554, 0.3264],
        [1.9428, 0.3264, 1.3823]])
>>> LD, pivots, info = torch.linalg.ldl_factor_ex(A)
>>> LD
tensor([[ 7.2079,  0.0000,  0.0000],
        [ 0.5884,  0.9595,  0.0000],
        [ 0.2695, -0.8513,  0.1633]])
>>> pivots
tensor([1, 2, 3], dtype=torch.int32)
>>> info
tensor(0, dtype=torch.int32)
```

Notes & Warnings

NOTE

Note When the inputs are on a CUDA device, this function synchronizes only when `check_errors= True`.

⚠ WARNING

Warning This function is “experimental” and it may change in a future PyTorch release.

Detailed Documentation

Documentation

This is a version of `ldl_factor()` that does not perform error checks unless `check_errors=True`. It also returns the `info` tensor returned by LAPACK's `sytrf`. `info` stores integer error codes from the backend library. A positive integer indicates the diagonal element of `DDD` that is zero. Division by 0 will occur if the result is used for solving a system of linear equations. `info` filled with zeros indicates that the factorization was successful. If `check_errors=True` and `info` contains positive integers, then a `RuntimeError` is thrown.

When the inputs are on a CUDA device, this function synchronizes only when `check_errors=True`.

This function is “experimental” and it may change in a future PyTorch release.

`A` (Tensor) – tensor of shape $(*, n, n)$ where $*$ is zero or more batch dimensions consisting of symmetric or Hermitian matrices.

`hermitian` (bool, optional) – whether to consider the input to be Hermitian or symmetric. For real-valued matrices, this switch has no effect. Default: `False`.

`check_errors` (bool, optional) – controls whether to check the content of `info` and raise an error if it is non-zero. Default: `False`.

`out` (tuple, optional) – tuple of three tensors to write the output to. Ignored if `None`. Default: `None`.

A named tuple (`LD`, `pivots`, `info`).

